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17

Why?

GUT COST, OVERCS,

GQMC

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fhe - quantum penguin

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Qubits

→ Vectors in 2D-plane → Complex space → Known as State space

$$\underline{|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}}$$

$$\underline{|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\alpha|^2 + |\beta|^2 = 1$$

↳ α, β can be complex.

Quantum Logic Gates

Quantum NOT Gate (X)

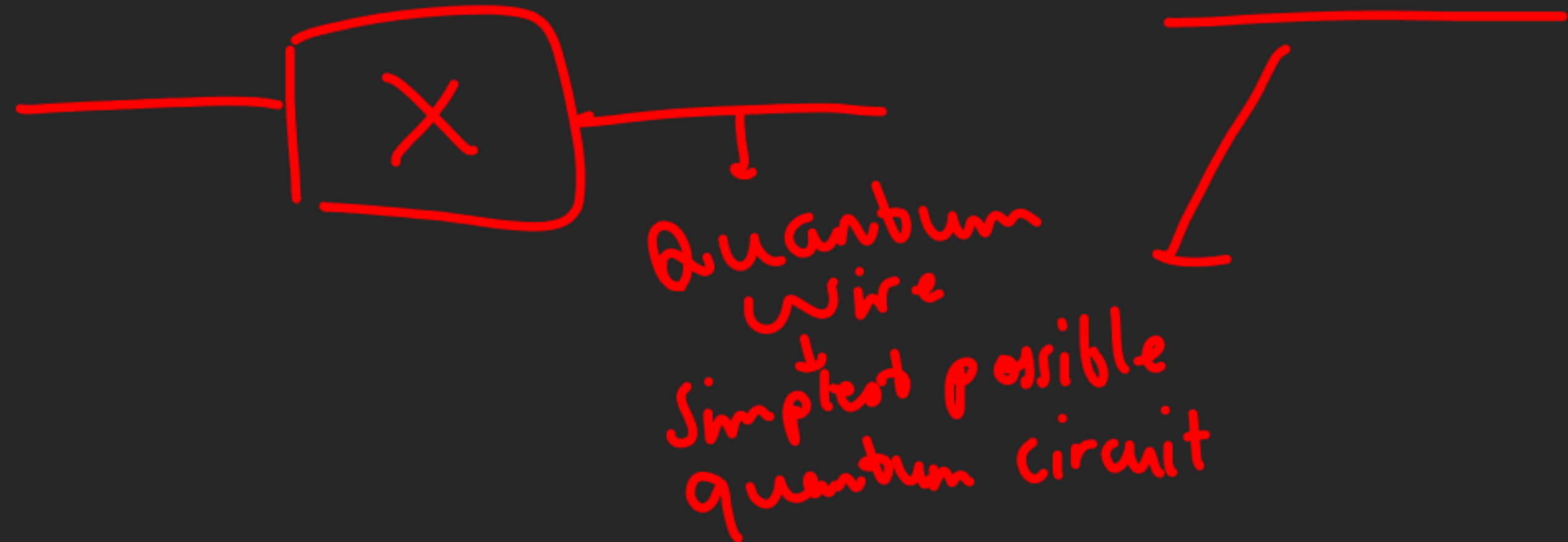
$$X|0\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$X|1\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

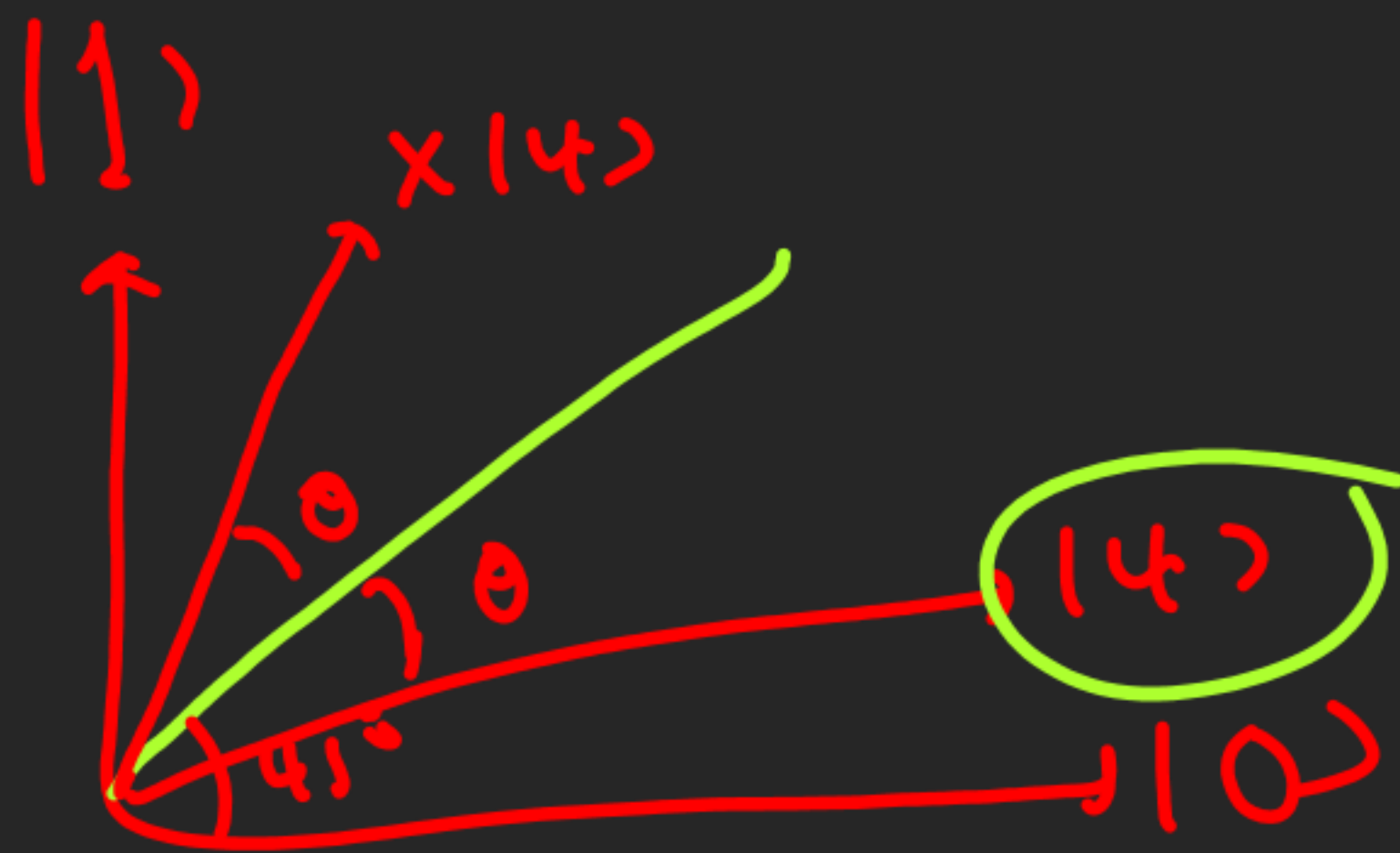
$$6 \times 6$$

$$X(\alpha|0\rangle + \beta|1\rangle) = \alpha|1\rangle + \beta|0\rangle$$





$$X \cdot X = I \rightarrow X^2 = I \rightarrow X = X^{-1}$$



— → Mirror

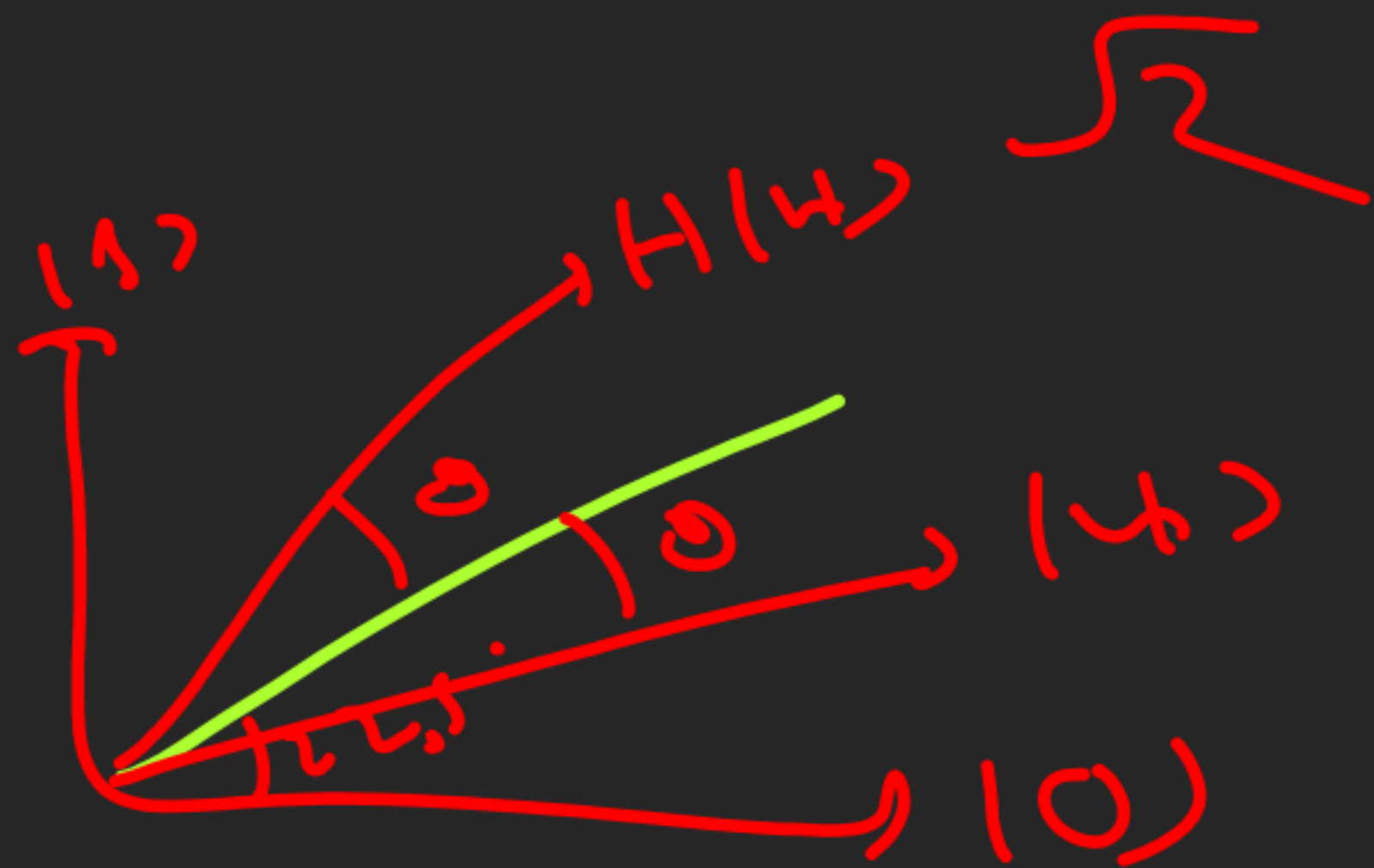
→ Neutrino → Bad Qubit → Interacts less, but Qubit is stable in comparison

↓
Generates less error

Hadamard Gate

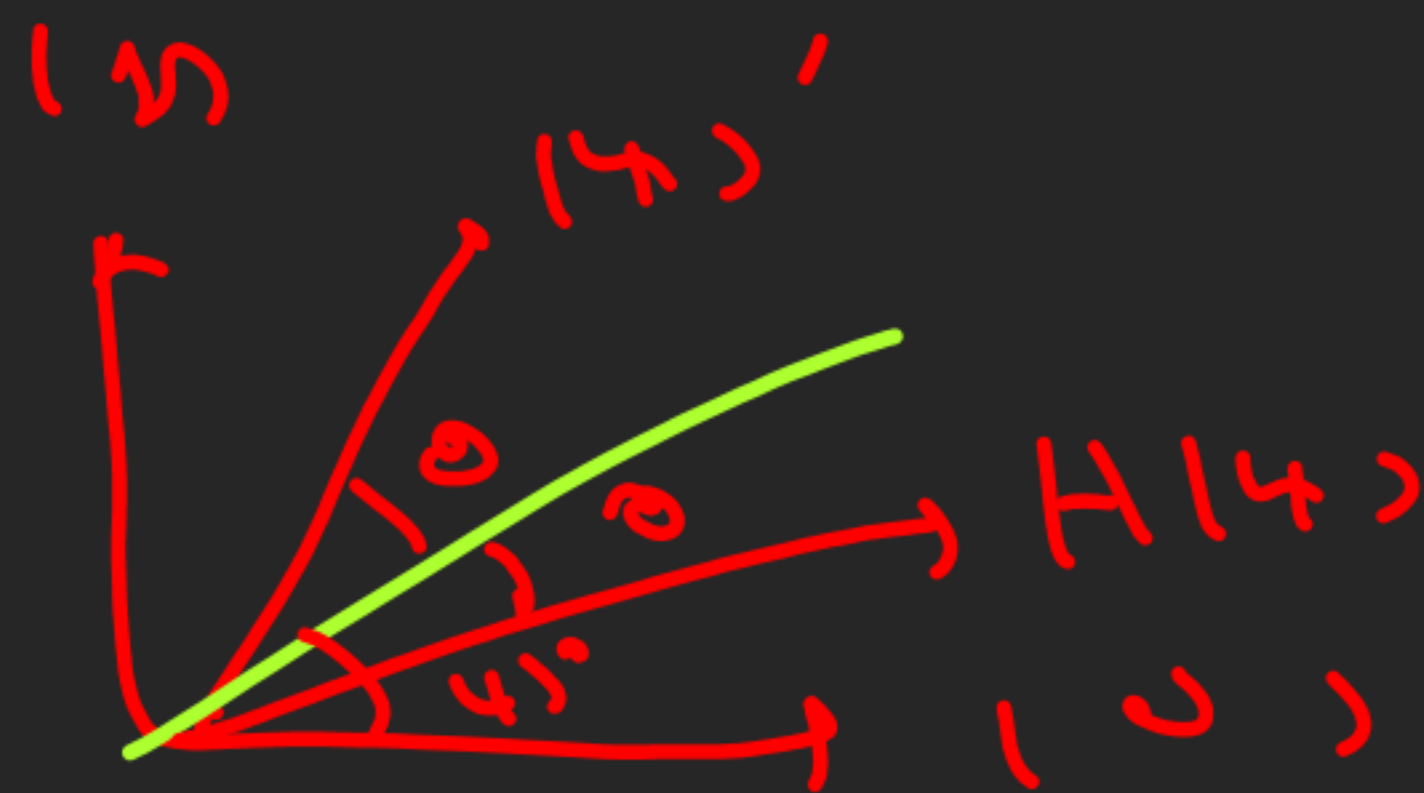
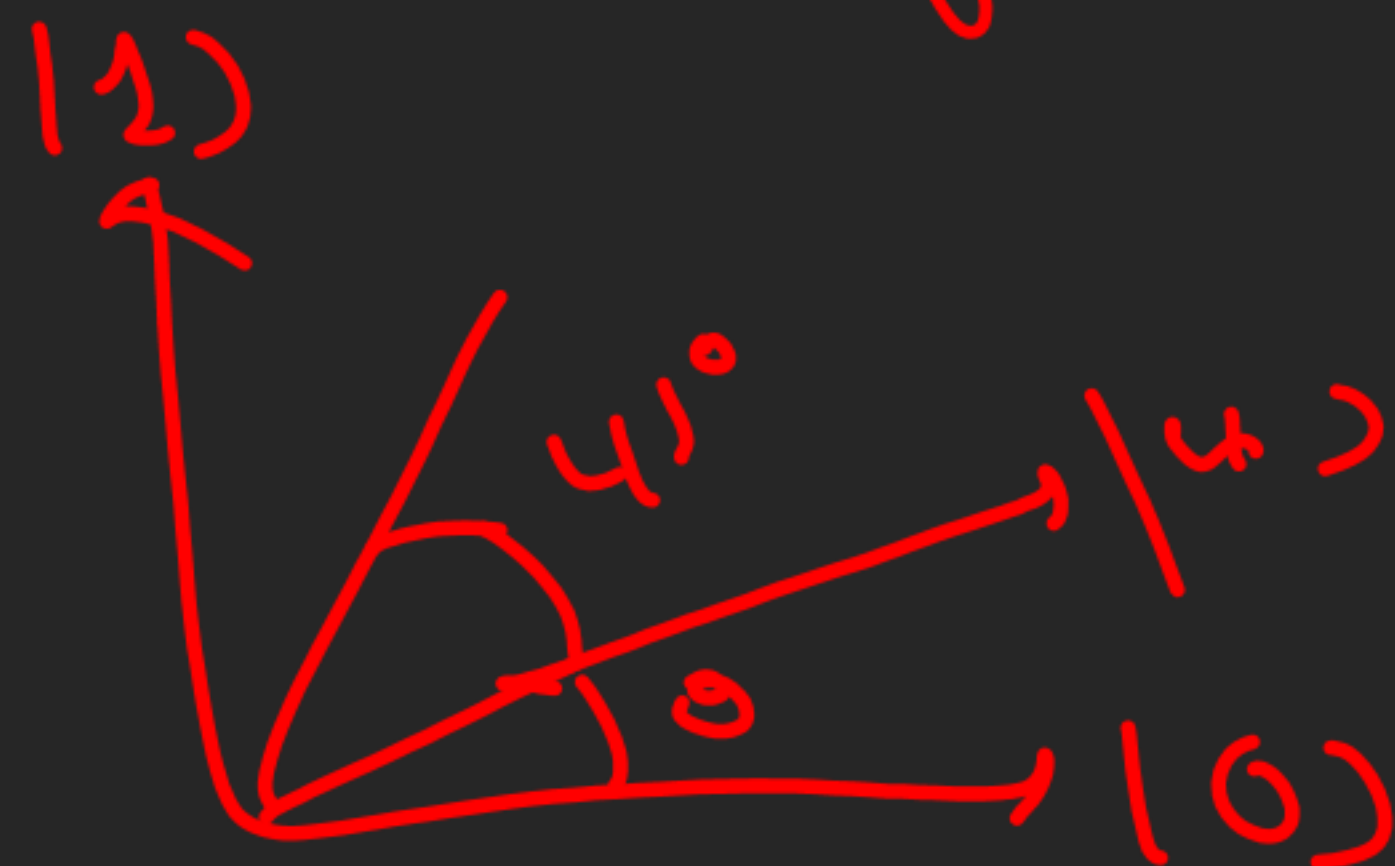
$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



↓
Mirror at angle 22.5°

Alternatively,

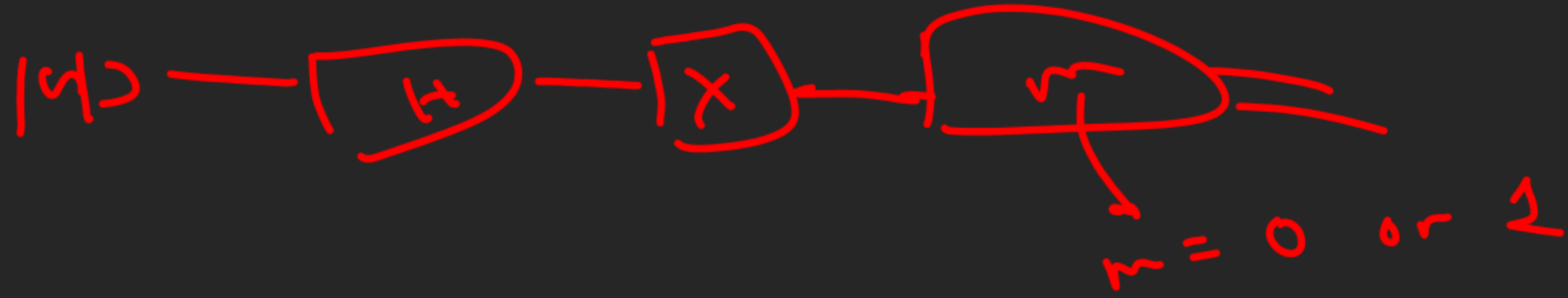


Measuring a Qubit

$\alpha|0\rangle + \beta|1\rangle$ $\xrightarrow{\text{observed}}$ $\textcircled{0}$ $\textcircled{1}$

$|0\rangle$ or $|1\rangle$
 $|\alpha|^2$ $|\beta|^2$

$\rightarrow$ Wave function collapses



General Single Qubit Gates

2x2 Unitary Matrix $\rightarrow U$
in general

$$UU^\dagger = I$$

$$|U| = 1$$

$U^\dagger \rightarrow$ Conjugate Transpose
of matrix

$$= (U^*)^T = (U^T)^*$$

Can only rotate,
not change length

$$\|U|4\rangle\| = \|4\rangle\|$$

Pauli Gates

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$X|0\rangle = |1\rangle$$

$$X|1\rangle = |0\rangle$$

σ_x / σ_1

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$Y|0\rangle = i|1\rangle$$

$$Y|1\rangle = -i|0\rangle$$

$$x, y, z (\propto |0\rangle + |1\rangle)$$

σ_y / σ_2

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = -|1\rangle$$

σ_z / σ_3

Ans.

Rotation Matrix

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

$$R_\theta^2 = R_{2\theta}$$

Bra-Ket / Bra-Ket Notation

$$|4\rangle = \begin{pmatrix} a \\ b \\ c \\ 1 \end{pmatrix}$$

Ket
Psi

$$(|4\rangle)^T = \langle 4| = (a^* \ b^* \ c^* \ \underline{\quad} \ 1^*)$$

Bra
Psi

$$(|4\rangle)^T = \langle 4|$$

$$\langle 4|^T = |4\rangle$$

$$||4\rangle||^2 = \langle 4|4\rangle$$

$$\begin{aligned} \langle 4|4\rangle &= a a^* + b b^* + c c^* + 1 \cdot 1^* \\ &= |a|^2 + |b|^2 + |c|^2 + 1 \end{aligned}$$

$$(AB)^T = B^T A^T$$

$$||M|4\rangle||^2 = \langle 4|M^T M|4\rangle$$

Another useful notation

$$\delta_{kj} = \begin{cases} 1 & \text{when } k=j \\ 0 & \text{when } k \neq j \end{cases}$$

→ Basically Identity Matrix

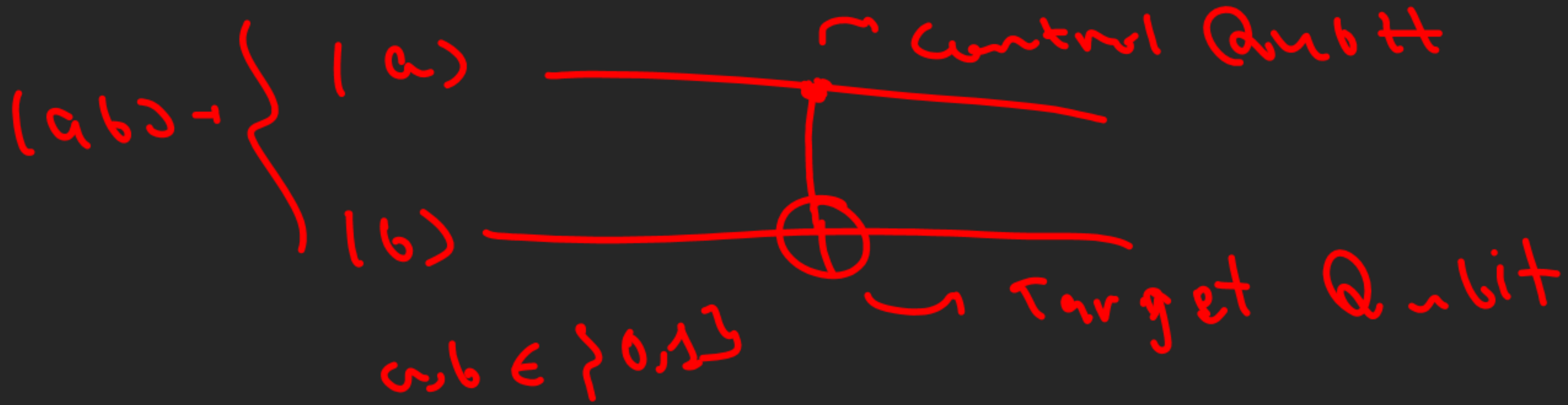
$$|e_j\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \rightarrow j\text{th row}$$

$$\langle e_k| = (0 \ 0 \ \underline{1} \ 0 \ 0)$$

kth column

$$M_{kj} = \langle e_k | M | e_j \rangle$$

CNOT Gate / Controlled-Not Gate



$|00\rangle \rightarrow |00\rangle$
 $|01\rangle \rightarrow |01\rangle$
 $|10\rangle \rightarrow |11\rangle$
 $|11\rangle \rightarrow |10\rangle$

→ If control qubit is 1
Target qubit is NOTed
or else it doesn't affect

$$|a, b\rangle \xrightarrow{\text{CNOT}} |a, b \oplus a\rangle$$

$$\text{CNOT}(\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle) \rightarrow$$

Ans.

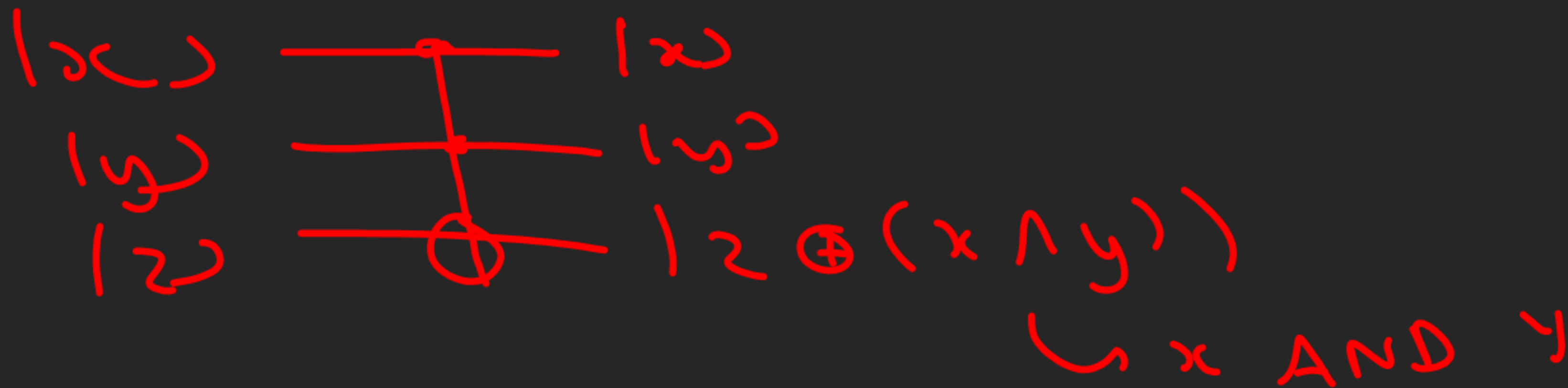


$$\rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



$$\rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

To folli: Gate



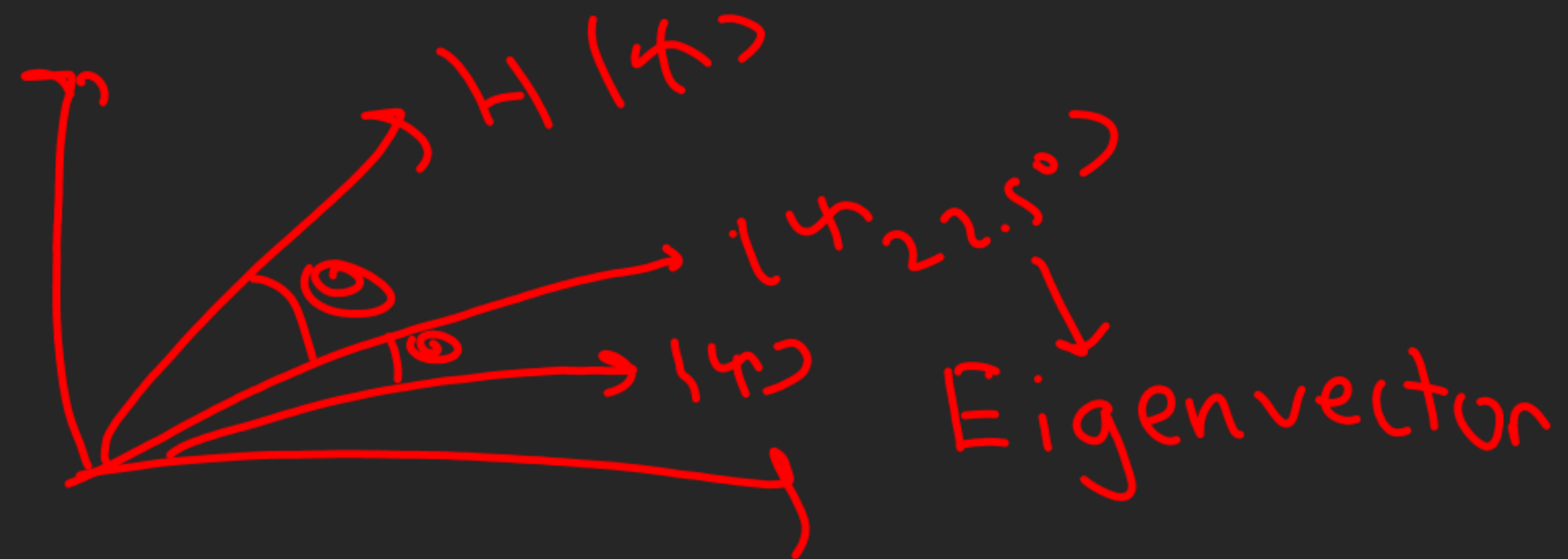
$z=0 \rightarrow$ AND Gate Function

$z=2 \rightarrow$ NAND Gate Function

$\neg(x \wedge y) \rightarrow (x \uparrow y)$
 $\downarrow$
 $x \text{ NAND } y$

Eigenvectors and Eigenvalues

$$U|4\rangle = \lambda|4\rangle$$



$$H|4_{22.5^\circ}\rangle = +|4_{22.5^\circ}\rangle$$

$$H|4_{202.5^\circ}\rangle = -|4_{202.5^\circ}\rangle$$

$$X|+\rangle = +|+\rangle$$

$$X|-\rangle = -|-\rangle$$